Deduce the World: How Far Can Reasoning Go?

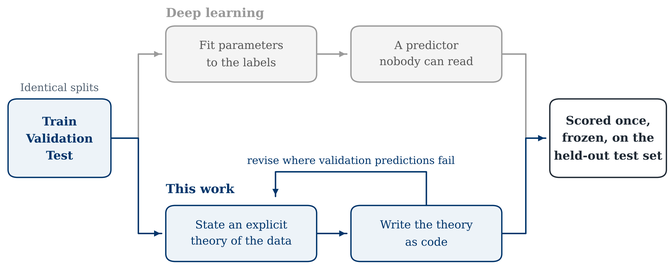
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Results & figures

1 Theories, not fitted predictors

Science deduces the world from first principles, revising its theories where their predictions fail. Deep learning does not. It fits a predictor to existing data, which binds it to that distribution.

We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.

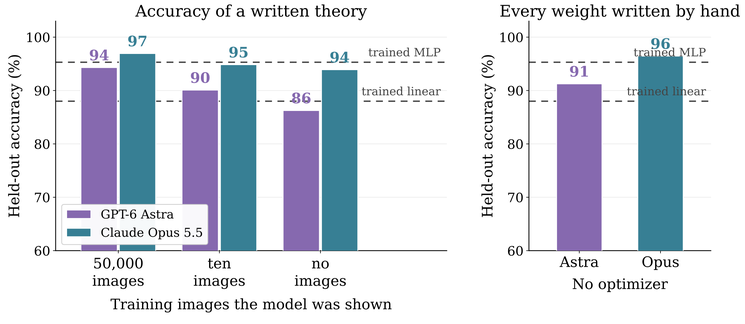


We keep the splits of deep learning and replace the learner. The model states a theory, writes it as code, and revises it against validation.

May	read papers and the web, measure the training data, inspect its own errors
May not	fit parameters to labels, search automatically over rules, store examples, see the test split

2 How much data a theory needs

MNIST, the standard handwritten-digit benchmark: 60,000 training and 10,000 test images of the digits 0 to 9.



Best of five reasoning efforts. Dashed lines are trained models on the same split.

> Given no training images at all, a theory of how digits are written classifies most of them.

Reasoning as initialization

A written theory is also a starting point. Networks initialized from written weights reach lower held-out error after the same training budget than networks initialized at random, and the gap widens with reasoning effort: error falls as $r^{-0.46}$ against $r^{-0.21}$.

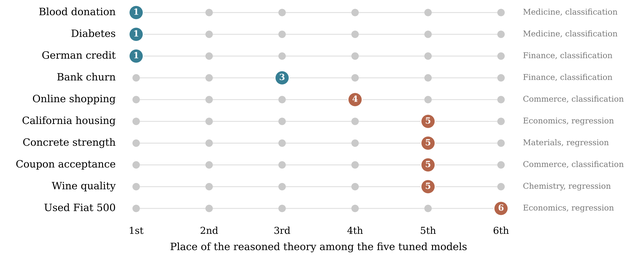
3 The same protocol, ten fields

Ten tasks from TabArena, a curated benchmark of real tabular prediction problems with fixed splits and tuned baselines.

What a theory looks like

Selected rules from one program on German credit, one of the ten tasks above. The points are judgments stated before measurement, not fitted odds ratios.

Checking account	none $+3$, at least 200 DM $+1$, negative -2
Credit history	critical account elsewhere $+2$, past delay -1
Loan duration	-1 for each twelve months beyond the first twelve
Output	$P(\text{good}) = 1/(1+e^{-(\text{score}-1)/2})$

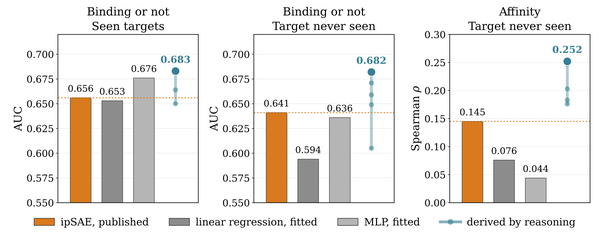


Each task ranks the reasoned theory against five tuned models on the same rows.

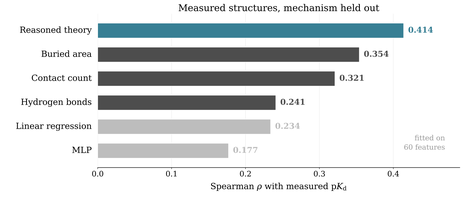
> Reasoned theories win on small clinical and credit tables, and trail boosted trees where the signal lies in feature interactions.

4 Deriving a scientific score

A structure predictor returns a complex and an estimate of its own error. Each session receives that matrix alone.



704 designed binders. The middle and right panels hold out an entire target.



One mechanism of association held out whole. Grey bars are fitted on sixty features.

> Every unfitted physical quantity beats both fitted models, and the reasoned rule beats all of them.

What a theory looks like

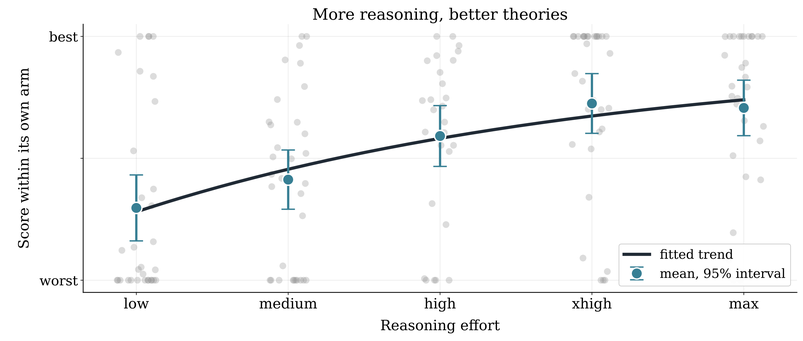
The chain one session stated before writing any code. Nothing in it is fitted.

Harmonic well	Equipartition sets the force constant to the thermal energy over the square of the fluctuation. The cross-chain error estimates it.
Stiffness is depth	For a well of fixed shape, curvature at the minimum is proportional to depth. A stiffer interface is a deeper one.
Divide out the predictor	The reported error inflates where the predictor modelled the chain poorly. Measuring it against the binder's own internal error removes that, and the units with it.
Count what binding confines	Association removes six rigid-body degrees of freedom. Each confined quadratic degree contributes half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The three is six halved. The reference affinity is an additive constant, so it cannot touch a rank correlation.

5 Scaling reasoning effort



Every arm of the project on one axis, each normalized to its own range. The shaded band is the interquartile range across all arms.

> More reasoning makes better theories where theory is what is missing, and makes none where the model already derives the answer at once.

6 Hypothesis

Deep learning dominates the domains that are data-rich and theory-poor. We hypothesize that as reasoning improves, language models will supply the first principles those domains lack, and that stated theories, rather than fitted predictors, will become how we model them.